BIOGRAPHICAL SKETCH

NIH-style layout, not the official NIH form. Data as of 5 November 2026.

Name	Dr Ramachandran Thiruvengadam
Position title	Associate Professor, Maternal and Child Health Programme, BRIC-THSTI
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A. Personal statement

I am a physician-scientist (M.B.,B.S., DCH, M.D. Biochemistry, Ph.D.) working on the prediction, diagnosis and prevention of adverse pregnancy outcomes. I am an investigator in the GARBH-Ini pregnancy cohort programme and lead cohort analyses on small vulnerable newborns and gestational weight gain. I coordinated the clinical component of the DBT Consortium for COVID-19 Research and I design and co-lead trials and AI-enabled imaging projects, including the ICMR GRASPP randomised trial. I teach epidemiology, clinical research and biostatistics to PhD and MSc students.

B. Professional preparation

Institution	Degree	Year
Translational Health Science and Technology Institute, Faridabad (affiliated to Jawaharlal Nehru University, New Delhi)	Ph.D. (Life Sciences - Maternal and Child Health)	2023
All India Institute of Medical Sciences (AIIMS), New Delhi	M.D. (Biochemistry)	2016
Institute of Child Health and Hospital for Children, Madras Medical College, The Tamil Nadu Dr MGR Medical University	Diploma in Child Health (DCH)	2009
Madurai Medical College, Madurai, The Tamil Nadu Dr MGR Medical University	M.B.,B.S.	2005

C. Appointments and positions

Institution	Position	From	To
BRIC-Translational Health Science and Technology Institute (THSTI), Faridabad	Associate Professor, Maternal and Child Health Programme	30 June 2025	Till date
Pondicherry Institute of Medical Sciences, Puducherry	Assistant Professor, Department of Biochemistry	May 2022	June 2025
THSTI, Faridabad	Principal Scientist - I	November 2021	May 2022
THSTI, Faridabad	Senior Research Scientist	July 2021	November 2021
THSTI, Faridabad	Scientist D	January 2020	June 2021
THSTI, Faridabad	Scientist C	January 2019	January 2020
THSTI, Faridabad	Early Career Medical Research Awardee	July 2016	January 2019
All India Institute of Medical Sciences, New Delhi	Junior Resident (M.D. Biochemistry)	2013	2016

D. Honors

- Award for a PhD student for the Best Published Paper of 2021, THSTI Foundation Day (DBT) (15 July 2022)
- Best oral presentation (second prize), Workshop on Air Pollution and Health Effects Research in India, CAPHER-India (21-23 October 2021)
- Best PhD oral presentation (second prize), National Science Day, THSTI (28 February 2021)
- Dr M.K. Bhan Group Award for the most impactful collaboration, THSTI (as a member of the clinical team of the DBT Consortium for COVID-19 Research) (2020)
- Early Career Medical Research Award, THSTI, DBT, Government of India (maternal and child health) (2017)

E. Contributions to science

1. Gestational age estimation and pregnancy dating. Within the GARBH-Ini pregnancy cohort I contributed to the development and external validation of Indian population-specific models for estimating gestational age in the second and third trimesters (Garbhini-GA2), and to a comparison of first-trimester dating methods and their effect on preterm birth classification. This work underlies two Indian patent applications, including an ultrasound image-based gestational age estimation system (GAUGE).

Gadekar VP, Damaraju N, Xavier A, et al; GARBH-Ini Study Group. Development and external validation of Indian population-specific Garbhini-GA2 model for estimating gestational age in second and third trimesters. *Lancet Reg Health Southeast Asia*. 2024;25:100362. doi:10.1016/j.lansea.2024.100362. PMID: 39021476

Vijayram R, Damaraju N, Xavier A, et al; GARBH-Ini Study Group. Comparison of first trimester dating methods for gestational age estimation and their implication on preterm birth classification in a North Indian cohort. *BMC Pregnancy Childbirth*. 2021;21(1):343. doi:10.1186/s12884-021-03807-4. PMID: 33931016

Ramasamy T, Varughese B, Singh M, et al. Post-natal gestational age assessment using targeted metabolites of neonatal heel prick and umbilical cord blood: a GARBH-Ini cohort study from North India. *J Glob Health*. 2024;14:04115. doi:10.7189/jogh.14.04115. PMID: 38968007

2. Small vulnerable newborns and gestational weight gain. I led the analysis of incidence and risk factors for small vulnerable newborns in the GARBH-Ini cohort and described gestational weight gain trajectories against global references, with related work on weight gain monitoring in low- and middle-income settings and on weight gain and adverse pregnancy outcomes across South Asia and sub-Saharan Africa.

Thiruvengadam R, Ayushi, Murugesan DR, et al; GARBH-Ini Study Team. Incidence of and risk factors for small vulnerable newborns in north India: a secondary analysis of a prospective pregnancy cohort. *Lancet Glob Health*. 2024;12(8):e1261-e1277. doi:10.1016/S2214-109X(24)00212-2. PMID: 39030058

Thiruvengadam R, Desiraju BK, Natchu UCM, et al; GARBH-Ini Study Team. Gestational weight gain trajectories in GARBH-Ini pregnancy cohort in North India and a comparative analysis with global references. *Eur J Clin Nutr*. 2022;76(6):855-862. doi:10.1038/s41430-021-01040-y. PMID: 34785811

Thiruvengadam R, Desiraju BK, Sachdev HS, Bhatnagar S. The challenges in gestational weight gain monitoring in low and middle income settings. *Eur J Clin Nutr*. 2023;77(7):764-765. doi:10.1038/s41430-023-01292-w. PMID: 37316558

Sebastião YV, Thiruvengadam R, Khanam R, et al. Association between gestational weight gain and adverse pregnancy outcomes: cohort analysis from South Asia and sub-Saharan Africa. *BMJ Public Health*. 2025;3(1):e000900. doi:10.1136/bmjph-2024-000900. PMID: 40433067

3. Biomarkers and multi-omics of preterm birth. I contributed clinical phenotyping and epidemiology to multi-omics studies of spontaneous preterm birth in GARBH-Ini, covering genetic variants, post-GWAS integration of multi-omics data, vaginal microbiome signatures and placental proteomics.

Bhattacharjee E, Thiruvengadam R, Varshney D, et al; GARBH-Ini Study Group. Deciphering mechanisms leading to preterm birth through post-GWAS integration of multi-omics data. *iScience*. 2025;28(11):113861. doi:10.1016/j.isci.2025.113861. PMID: 41362759

Bhattacharjee E, Thiruvengadam R, Ayushi, et al; GARBH-Ini Team. Genetic variants associated with spontaneous preterm birth in women from India: a prospective cohort study. *Lancet Reg Health Southeast Asia*. 2023;14:100190. doi:10.1016/j.lansea.2023.100190. PMID: 37492417

Talukdar D, Sarkar M, Ahrodia T, et al. Previse preterm birth in early pregnancy through vaginal microbiome signatures using metagenomics and dipstick assays. *iScience*. 2024;27(11):111238. doi:10.1016/j.isci.2024.111238. PMID: 39569373

Kharbanda N, Biswas A, Tiwari A, et al. Placental proteomics reveals an elevated level of aldo-keto reductase 1-B1, highlighting its potential role in spontaneous preterm birth. *J Proteome Res*. 2025;24(2):612-623. doi:10.1021/acs.jproteome.4c00698. PMID: 39762117

4. COVID-19 vaccine effectiveness and immunity. As clinical coordinator of the DBT Consortium for COVID-19 Research I led the test-negative evaluation of ChAdOx1 nCoV-19 effectiveness during the Delta surge in India with a mechanistic immune-response study, and the clinical and experimental comparison of Omicron and Delta, and I contributed to longitudinal serology of infected individuals.

Thiruvengadam R, Awasthi A, Medigeshi G, et al; DBT India Consortium for COVID-19 Research. Effectiveness of ChAdOx1 nCoV-19 vaccine against SARS-CoV-2 infection during the delta (B.1.617.2) variant surge in India: a test-negative, case-control study and a mechanistic study of post-vaccination immune responses. *Lancet Infect Dis*. 2022;22(4):473-482. doi:10.1016/S1473-3099(21)00680-0. PMID: 34838183

Thiruvengadam R, Rizvi ZA, Raghavan S, et al. Clinical and experimental evidence suggest omicron variant of SARS-CoV-2 is inherently less pathogenic than delta variant independent of previous immunity. *Eur J Med Res*. 2023;28(1):421. doi:10.1186/s40001-023-01373-3. PMID: 37821945

Thiruvengadam R, Chattopadhyay S, Mehdi F, et al; DBT India Consortium for COVID-19 Research. Longitudinal serology of SARS-CoV-2-infected individuals in India: a prospective cohort study. *Am J Trop Med Hyg*. 2021;105(1):66-72. doi:10.4269/ajtmh.21-0164. PMID: 34003792

Thiruvengadam R, Binayke A, Awasthi A. SARS-CoV-2 delta variant: a persistent threat to the effectiveness of vaccines. *Lancet Infect Dis*. 2022;22(3):301-302. doi:10.1016/S1473-3099(21)00697-6. PMID: 34826382

5. AI-enabled obstetric ultrasound and clinical prediction models. I contribute clinical expertise to AI methods for obstetric ultrasound sweeps and to clinical prediction models for pregnancy outcomes, including reviews of spontaneous preterm birth prediction and gestational diabetes prediction models.

Gleed AD, Mishra D, Self A, et al. Statistical characterisation of fetal anatomy in simple obstetric ultrasound video sweeps. *Ultrasound Med Biol*. 2024;50(7):985-993. doi:10.1016/j.ultrasmedbio.2024.03.006. PMID: 38692940

Self A, Chen Q, Desiraju BK, et al; CALOPUS Study Group. Developing clinical artificial intelligence for obstetric ultrasound to improve access in underserved regions: protocol for a Computer-Assisted Low-Cost Point-of-Care UltraSound (CALOPUS) study. *JMIR Res Protoc*. 2022;11(9):e37374. doi:10.2196/37374. PMID: 36048518

Chanian R, Thiruvengadam R, Desiraju BK. Prediction of spontaneous preterm birth in the second trimester: current models, limitations, and the way forward. *Indian J Pediatr*. 2026;93(8):852-858. doi:10.1007/s12098-026-06030-1. PMID: 41729215

Dinakaran A, Thiruvengadam R, Srinivasan AR, et al. Prediction models for gestational diabetes mellitus: diagnostic utility of clinical and biochemical markers. *Indian J Endocrinol Metab*. 2025;29(3):295-302. doi:10.4103/ijem.ijem_497_24. PMID: 40688618

F. Synergistic activities

- Expert member, Project Selection Committee (Epidemiology), ICMR intramural funding, 2024 onwards.
- Academic Editor, PLOS Global Public Health, and Associate Editor, Frontiers in Pediatrics, 2023 onwards.
- Member, Technical Advisory Group, Obesity Individual Participant Data Meta-analysis, ICMR (Division of Reproductive, Child Health and Nutrition), 2026 onwards.
- Faculty, MSc Clinical Research programme (clinical trials specialisation), THSTI: Epidemiology module (2023 onwards).
- Faculty and co-facilitator, EpiCap, a capacity-building programme for epidemiology teachers in LMICs (ICMR-NIE and University of Bergen) (2023 onwards).

G. Patents

- Garbh-Ini Ultrasound Image-based Gestational Age Estimation System (GAUGE). Indian application 202411027837 (filed 4 april 2024; awarded).
- A population-specific gestational dating system and a method thereof. Indian application 202341061748 (filed 13 september 2023).
- Potently neutralising novel human monoclonal antibodies to SARS-CoV-2 (COVID-19). Indian application 202111038519 (filed 25 august 2021; awarded).
- Immunoassay for detection of Coronaviridae. Indian application 202011039696 (filed 14 september 2020; awarded).